14.2. THE ACTINIDE ELEMENTS

14.2.1. Actinium, element 89

Actinium was discovered in 1899 by Debierne in residues from the work of the Curies. It was named after the Greek word for beam or ray, aktis. Actinium occurs in nature as a member of the long natural decay series, except the one beginning with ^{238}U, see Fig. 1.2. Being the first of them, it has given name to the actinide series of elements. Chemically actinium behaves very similar to the lanthanides and especially as lanthanum.

14.2.1.1. Isotopes

Actinium only occurs in nature as the isotope ^{227}Ac ($t_{½}$ 21.77 y), which is a member of the long decay series starting with ^{235}U. This is also the most stable actinium isotope.

14.2.1.2. Occurrence and production

Due to its very low concentration, actinium is almost never produced from the ^{235}U decay series. However, it has been produced in macro amounts by neuron irradiation of ^{226}Ra:

$$^{226}_{88}\mathrm{Ra} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{227}_{88}\mathrm{Ra} \rightarrow {}^{227}_{89}\mathrm{Ac} + \beta^{-} \tag{14.2}$$

14.2.1.3. Uses

Actinium has been used in radioactive n-sources made by compressing an intimate mixture of ^{227}Ac and beryllium. The neutrons are produced when the α-particles from actinium hit atoms of ^{9}Be. It was earlier suggested that actinium compounds might be used in radiation therapy of cancers, however, better alternatives now exist.

14.2.2. Thorium, element 90

Thorium was discovered by Berzelius in 1828 from a sample of what is now called thorite ($ThSiO_4$) and named by him after the nordic god of thunder, Thor. As a metal it is silvery white but with exposure to air it tarnishes slowly becoming gray and finally black.

14.2.2.1. Isotopes

Thorium occurs in nature as the isotopes ^{227,228,229,230,231,231,232,234}Th, most of them short lived. Natural thorium consists almost to 100% by weight of the very long-lived isotope ^{232}Th ($t_{½}$ 1.405×10^{10} y), which is the parent nuclide of the thorium decay series. The specific radioactivity for thorium is lower than that of uranium, and it is often treated as a non-radioactive element. For radioactive tracer studies, the nuclide ^{234}Th ($t_{½}$ 24.1 d) is usually separated from natural uranium.

14.2.2.2. Occurrence and production

Thorium is somewhat more common in nature than uranium, with an average content in the earth's crust of 10 ppm (by comparison the average abundance of lead is about 16 ppm in the earth's crust). In minerals thorium mainly exist as oxides but also silicates and phosphates exist. The content of thorium in sea water is $<0.5 \times 10^{-3}$ g/m^3, which is lower than that of uranium because thorium only exist as the low solubility tertavalent form while in the case of uranium oxidation states with higher solubility, e.g. (+VI) also exist and thus increase the total solubility.

The most common thorium mineral is monazite, a golden brown rare earth phosphate containing 1–15% ThO_2 and usually 0.1–1% U_3O_8. It is also found in small amounts in granite and gneiss. The largest deposits of monazite are found in India, Egypt, South Africa, the USA, and Canada, with 200–400 kton ThO_2 in each country. The size of natural resources are defined in terms of ore reserves which can be economically processed at the current market price. Thus, the total reserves at the 2005 market price was estimated by IAEA to ~2.6 Mt Th.

Because thorium often occurs with other valuable metals (in addition to the lanthanides) such as niobium, uranium and zirconium, it can be produced as a byproduct.

The following procedure is used for producing thorium from monazite sand. The sand is digested with hot concentrated alkali which converts the oxide to hydroxide. The filtered hydroxide is dissolved in hydrochloric acid and the pH adjusted between 5 and 6, which precipitates the thorium hydroxide but not the main fraction of lanthanide elements. The thorium hydroxide is dissolved in nitric acid and selectively extracted with methyl isobutyl ketone or tributyl phosphate in kerosene. This gives a rather pure organic solution of $Th(NO_3)_4$. The thorium is stripped from the organic phase by washing with alkali solution.

For the case of production of ^{234}Th it is conveniently done from an ether solution containing uranyl nitrate. Upon contact with some ml of water the thorium is stripped to the water phase. This waster phase is then run through a cation exchange column, washed with 6M HCl and then eluted with 0.5M oxalic acid. The oxalic acid is removed from the solution by wet combustion and the remaining solution is once again passed through a cation exchange column and the procedure is repeated. In the last step the remaining water phase is diluted in the desired acid and the stock solution is ready.

14.2.2.3. Uses

Thorium metal is used as electrode material in gas discharge lamps, and as getter for absorption of rest gases in high vacuum technique. ThO_2 (melting point 3300°C) is highly refractory and used for high temperature furnace linings and in ultrasonic very high temperature thermometers. It is also alloyed with magnesium to improve the high temperature performance Thorium salts are of little practical use. Because Th^{4+} is a stable tetravalent ion with properties very similar to the tetravalent actinides, Th^{4+} is

often used as an analog for the An(IV) ions; most commonly used isotopes are ^{228}Th ($t_{½}$ 1.91 y), ^{230}Th ($t_{½}$ 7.54×10^4 y) and ^{234}Th ($t_{½}$ 24.5 d), which can be isolated from old ^{232}Th or ^{238}U, see Fig. 1.2. This use of thorium as an analogue for tetravalent actinides is however rather questionable since thorium is considerably larger and do not have any 5f electrons. The size difference matters for so called hard ligands like oxygen while the 5f electrons matter for softer ligands like nitrogen. For hard ligand e.g. zirconium is a better analogue for e.g. plutonium (IV) Thorium may become important as fuel for nuclear reactors due to its relative abundance and the fact that no plutonium is created during reactor operation (Ch. 20). On the other hand, the ^{233}Pa → ^{233}U decay in spent fuel may be used to produce pure ^{233}U, which can replace ^{239}Pu in explosive nuclear devices.

14.2.3. Protactinium, element 91

Protactinium (the parent of actinium) was first discovered in 1913 by Fajans and Göhring and initially named brevium due to the short half-life of the isotope used in the discovery, ^{234}Pa ($t_{½}$ 6.75 h). The elements existence was confirmed in 1918 when two groups independently discovered the longer lives isotope ^{231}Pa. These groups were Otto Hahn and Lise Meitner from Germany and Frederick Soddy and John Cranston from Great Britain. The name was then changed to protactinium as it was recognised as a parent of actinium. It is considered as one of the rarest naturally occurring element.

14.2.3.1. Isotopes

Protactinium occur in nature as the isotopes ^{231}Pa ($t_{½}$ 32760 y), ^{234m}Pa ($t_{½}$ 1.17 m), and ^{234}Pa ($t_{½}$ 6.75 h) − all members of long-lived natural decay chains, see Fig. 1.2. All known isotopes of protactinium are shown in Fig. 14.1.

14.2.3.2. Occurrence and production

Due to its long half-life in comparison with the other isotopes, ^{231}Pa is the dominant isotope in nature. It can be produced in macroscopic amounts from old uranium minerals and sometimes from wastes from U-production and purification. Most of the ^{231}Pa available today stems from a separation project during the early 1960:s where 125 g of ^{231}Pa was recovered from the waste from a defunct U-purification process based on ether extraction. Since then, most of that material has been lost to various wastes.

The shorter lived ^{233}Pa is produced by n-irradiation of ^{232}Th in a nuclear reactor.

$$^{232}_{90}\mathrm{Th} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{233}_{90}\mathrm{Th} \rightarrow {}^{233}_{81}\mathrm{Pa} + \beta^{-} \qquad (14.3)$$

By dissolution of the irradiated material in nitric acid, dilution to about 1 M and sorption of the ^{233}Pa on a MnO_2-filled column, delay for shorter-lived Pa-isotopes to decay, washing and elution by a dilute fluoride solution, ^{233}Pa can be produced.

14.2.3.3. Uses

Long-lived ^{231}Pa has no general use today. The main use of ^{233}Pa is as mother for isotopically pure ^{233}U. It is also an important intermediate in the fuel chain of thermal thorium reactors. Pa is one among the most toxic elements – about 2.5×10^8 times more toxic than HCN.

14.2.4. Uranium, element 92

Uranium was discovered in 1879 by Klaproth, who named it after the newly discovered planet Uranus. The specific radioactivity of natural uranium makes it a weak radiological hazard (see Ch. 15). It is also chemically toxic and precautions should be taken against inhaling uranium dust for which the threshold limit is 0.20 mg/m^3 air (about the same as for lead).

14.2.4.1. Isotopes

Natural uranium consists of 3 isotopes, ^{234}U, ^{235}U and ^{238}U, members of the natural decay series discussed in §1.3 and shown in Fig. 1.2. Uranium is an important raw material for nuclear energy production (see Ch. 19, 20 and 21). Fig. 14.1 shows all known uranium isotopes and some of their nuclear properties. A mass of nearly pure ^{235}U above a certain critical amount can undergo a nuclear explosion if suddenly hit by a rather high n-flux. The high initial n-flux requirement is due to the generation time of the fission neutrons and the limited time available until a supercritical amount is dispersed by the heat generated. This is further discussed in Chapter 19.

14.2.4.2. Occurrence, resources and production capacity

Uranium appears in a large number of minerals (at least 60 are known). The earth's crust contains 3–4 ppm U, which makes it about as abundant as arsenic or boron. Uranium is found at this relative concentration in the large granitic rock bodies formed by slow cooling of the magma about 1.7–2.5 eons ago (1 eon = 10^9 y = 1 billion years). It is also found in younger rocks at higher concentrations ("ore bodies").

Geochemists now begin to understand how these ore bodies were formed. They are usually located downstream from mountain ranges. As the atmosphere became oxidizing about 1.7 eons ago, rain penetrated into rock fractures and pores, bringing the uranium to its hexavalent state and dissolving it as an anionic complex (likely as a carbonate, $UO_2(CO_3)_n^{2-2n}$, or as a sulphate complex, $UO_2(SO_4)_n^{2-2n}$), often at elevated temperatures. As the water and the dissolved uranium migrated downstream, regions of reducing material were encountered, either inorganic (e.g. pyrite) or organic (e.g. humic) matter, which caused reduction to U(IV). Since many U(IV) compounds are insoluble in the natural environment, the uranium precipitated, possibly as the sulfide or, more likely, as

the hydroxide. Many of these original uranium precipitates were later covered by sedimentary material.

In most minerals uranium is in the tetravalent state. The most important one is uraninite (UO_{2+x}, $x = 0.01$ to 0.25), in which the uranium concentration is 50–90%; it is found in Western Europe, Central Africa (e.g. Katanga, Gabon), Canada (e.g. Cigar Lake) and Australia (e.g. Koongara). In the USA and Russia carnotite (a K + U vanadate) is the most important mineral and contains 54% uranium. In the high grade ores the mineral is mixed with other minerals so the average uranium concentration in the crushed ore is much less: e.g. $\leq$0.5% on the Colorado Plateau. The world's largest known uranium deposit is located at the Olympic Dam Mine in South Australia. Uranium is also often found in lower concentration, of the order of 0.01–0.03%, in association with other valuable minerals such as apatite, shale, peat, etc.

At a price of ~$130 per kg of U, the known (total of measured, indicated and inferred) world reserve of uranium which could be recovered economically at that price was estimated to be 4.7 Mtons. Sea water contains some 4 500 Mtons U but it is uncertain if U can be recovered economically from this huge resource.

Uranium production in 2010 was ~50.6 kton. Uranium producers are given in Table 14.1 by country.

Table 14.1 Uranium production by country in 2010

Country	(t U)	%
Kazakhstan	17803	33.18
Canada	9783	18.23
Australia	5900	10.99
Namibia	4496	8.378
Niger	4198	7.823
Russia	3562	6.638
Uzbekistan	2400	4.472
USA	1660	3.093
Ukraine	850	1.584
China	827	1.541
Malawi	670	1.249
South Africa	583	1.086
India	400	0.745
Czech Rep.	254	0.473
Brazil	148	0.276
Romania	77	0.143
Pakistan	45	0.084
France	7	0.013
TOTAL	53663	100

(data from WNA)

14.2.4.3. Production techniques

Uranium ores differ widely in composition, containing a variety of other elements which must be removed. As a result the production methods differ considerably depending on the particular ore to be processed although in every case very selective processes must be used. The following is a common scheme.

The ore is mined in open pits or underground. The ore is crushed and concentrated through flotation. If the uranium is in the tetravalent state it is oxidized in piles by air, sometimes with the aid of bacteria. The material is subsequently treated with sulfuric acid which dissolves the uranium as the sulphate complex, $UO_2(SO_4)_2^{2-}$ (the feed). This complex can be selectively removed from the aqueous solution by means of anion exchange resins or, more commonly, by extraction into an organic solvent (solvent extraction). In the latter case an extractant (i.e. an organic compound with ability to form a U-organic complex soluble in organic solvents) dissolved in kerosene is used; depending on the aqueous feed composition, various extractants have been applied. A typical flow-sheet is illustrated in Figure 14.2, and the basic chemical principles in Table 14.2. The final product is commonly ammonium diuranate, which is referred to as yellow cake; it contains 65—70% U. It is so free of radioactive uranium daughters that it can be handled safely in drums without any radiation protection. Yellow cake is further purified in order to obtain a final pure product of U_3O_8, usually better than 99.98% pure; its content of neutron poisons (nuclides which have high capture cross-sections for neutrons such as B, Cd, Dy, is less than 0.00002%, see Ch. 19).

..A different mining technique is the so called in situ leaching which is used all over the world, e.g. the Beverley mine in Australia. The basic idea is that a leaching solution is injected into the ore body in one place and then the loaded phase is pumped out at a different location (relatively close by). The site is surrounded by monitoring wells to make sure that no leach solution leaves the mining area. After the finishing of the mining activities ground water is pumped back in the holes and the mine is closed. Despite this seemingly simple remediation there are few in situ leaching sites where no active remediation was made leaving the ground heavily polluted with acidic uranium containing water. This problem was clearly shown in Saxonia in the former DDR where large amounts of money now have been spent on remediation since the German unification in 1990.

Heating yellow cake in hydrogen gas produces UO_2 which in sintered form is used as nuclear reactor fuel. If the UO_2 is exposed to F_2, UF_4 is obtained ("green salt"), which in a thermite process with calcium metal yields metallic uranium. The metal is slowly oxidized in air at room temperature; the metal powder is very reactive at elevated temperatures and can then be used to remove practically all impurities from rare gases.

The chemistry of aqueous uranium is discussed in §14.3 together with the chemistry of the other actinides. Production of reactor fuel and reprocessing is described in Chapter 21.

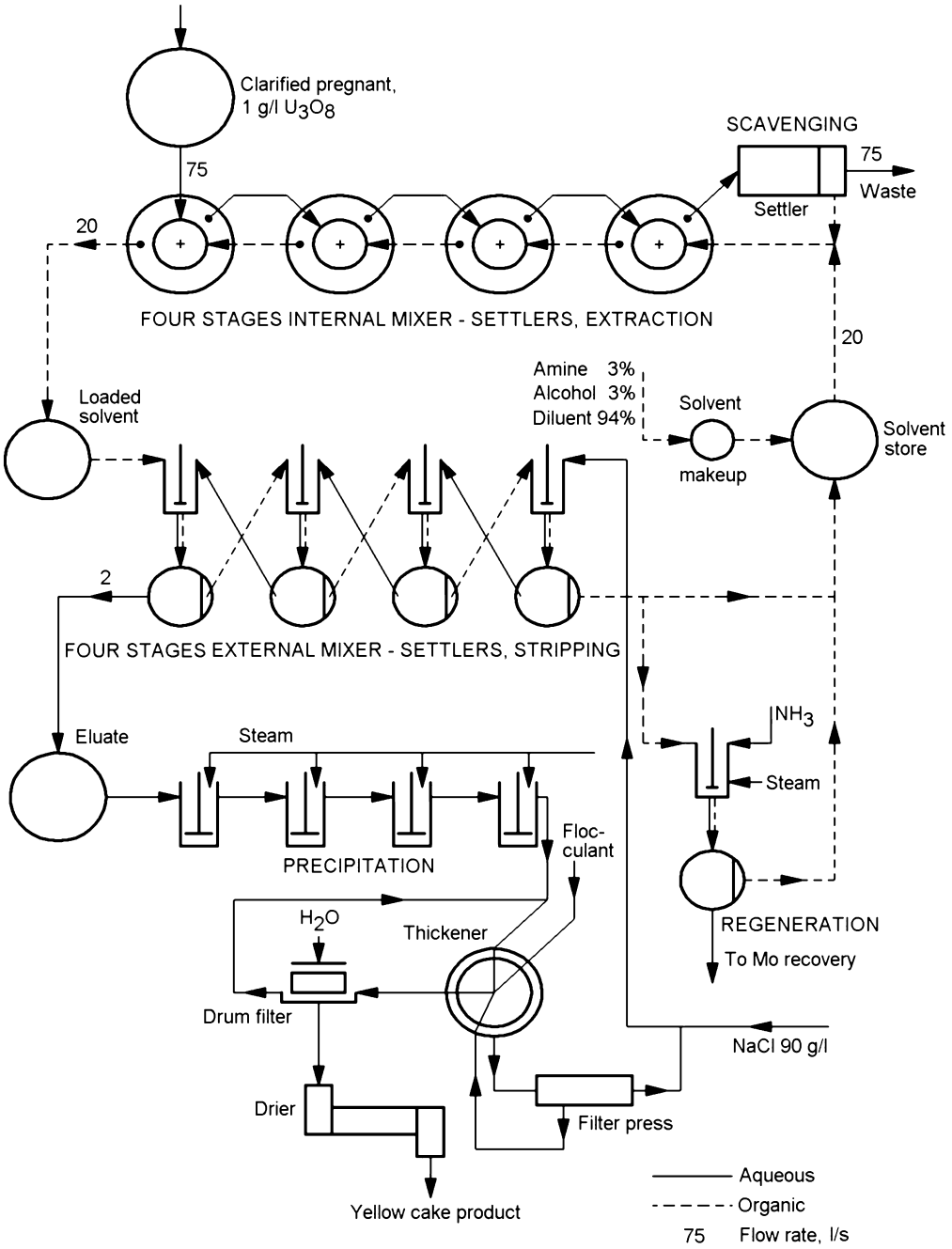


Figure 14.2 Amine extraction circuit, Kerr-McGee Corp., Grants Mill, New Mexico. Each S-X bettery consists of 4 mixer-settlers. Solid lines are aquous, and dashed lines are organic flow. Numbers are flow rates (l/s).

Table 14.2 Chemical bases for the Ames (amine) and Dapex (HDEHP) processes for uranium recovery from sulfuric acid leach liquors

	AMEX	DAPEX
Extraction	$UO_2^{2+} + SO_4^2 + 2(R_3NH)_2SO_4 \equiv (R_3NH)_4UO_2(SO_4)_3$ R_3N = Alamine 336 or Adogen 364 (trialkylamines with alkyl groups with 8–10 carbons)	$UO_2^{2+} + 2(HDEHP)_2 \equiv UO_2(HDEHP.DEHP)_2 + 2H^+$
Stripping	**Acidic stripping** $(R_3NH)_4UO_2(SO_4)_3 + 4HX \equiv 4R_3NHX + UO_2^{2+} + 3HSO_4 + H^+$ HX = HCl or HNO_3 **Neutral stripping** $(R_3NH)_4UO_2(SO_4)_3 + (NH_4)_2SO_4 + 4\ NH_3 \equiv 4R_3N + UO_2(SO_4)_3^4 + 6NH_4^+ + SO_4^2$ **Alkaline stripping** $(R_3NH)_4UO_2(SO_4)_3 + 7Na_2CO_3 \equiv 4R_3N + UO_2(CO_3)_3^4 + 4HCO_3 + 3SO_4^{2-} + 14Na^+$	**Alkaline stripping** $UO_2(HDEHP.DEHP)_2 + 4Na_2CO_3 \equiv UO_2(CO_3)_3^4 + 4NaDEHP + H_2O + CO_2 [+ 4Na^+$ (TBP must be added to avoid the third phase formation owing to low organic phase solubility of NaDEHP.) **Acidic stripping** $UO_2(HDEHP.DEHP)_2 + 2H^+ \equiv 2(HDEHP)_2 + UO_2^{2+}$

(From Musicas and Schulz.)

14.2.4.4. Production wastes

The milling operation produces tailings consisting of fine to coarse particles in a water slurry containing most of the radioactive uranium decay products, of which radium is the most hazardous. As most tailings are not highly radioactive (e.g. from mining of low grade ores) they are dumped on outside the plant; leach water from the dumps will then, combined with mine water, enter local streams. In dry areas, dusts from the tailings may spread by winds.

This is the main picture and explains the high dose commitment values in Table 22.1. However, many health authorities now require conditioning of the tailings to reduce harmful effects to the environment: recycling of waste water, precipitation and removal of radium from solution, and neutralization to precipitate heavy metals. The slurry is usually transported to an impoundment basin where the solid particles settle out, and the effluent is treated for removal of activities before discharge into a settling pond. Treatment ponds may contain low permeability liners where appropriate to control seepage. Barium chloride may be added to the tailings to precipitate dissolved ^{226}Ra as Ra–Ba sulfate. Lime and limestone may be added to the tailings to raise pH.

The final dry waste is stored either on surface or in shallow basins, though underground storage also has been practiced (abandoned mines). In the former case, the tailings are covered by up to 3 meters of earth fill to restrict erosion, and some water tight

material to protect it against rain. In some cases (e.g. Sweden) the land produced on top of mine waste and tailings have been reclaimed for farming.

14.2.4.5. Uses

The main use of uranium is as nuclear fuel, see Chapters 19–21. Depleted uranium from isotope separation plants is also used as metallic slugs in special kinds of armorpiercing ammunition. It has also been used as heavy ballast in some extreme racing yachts, but this has sometimes caused political problems. Depleted uranium has also been used as movable balance weights in large commercial aircraft. With the deployment of fast nuclear reactors, depleted uranium will probably find a new use as fertile reactor fuel. Uranium was previously also used as a commodity chemical. Due to its classification as a nuclear material ownership is now restricted in most parts of the world. Previously uranium was also used rather extensively for colouring of glass to a clear green colour which becomes intense when illuminated with UV light.

14.2.5. Neptunium, element 93

Early in 1940 McMillan and Abelson in the United States synthesized and identified a new element with atomic number 93 by bombarding uranium with slow neutrons. They named it neptunium after the planet Neptune. The reaction used in the synthesis was

$$^{238}_{92}\mathrm{U}(\mathrm{n},\gamma)^{239}_{92}\mathrm{U} \xrightarrow[23\ \mathrm{min}]{\beta^-} {}^{239}_{93}\mathrm{Np} \xrightarrow[2.3\ \mathrm{d}]{\beta^-} \tag{14.4}$$

(cf. Fig. 14.1). The experimental recoil technique for separating the fission and neutron capture products from the target material is described in §17.6.1. Chemical experiments showed that the product had a 2.3 d half-life and could be reduced by SO_2 to a lower valency state (presumably +4), which could be precipitated as a fluoride (carrier: LaF_3); this distinguished the element from uranium. In its oxidized state (using BrO as oxidant) it showed the same chemistry as hexavalent uranium. Because no fission product is expected to have this behavior, the assumption that the element with the 2.3 d half-life was a transuranic element was verified.

14.2.5.1. Isotopes

The longest lived isotope is ^{237}Np with $t_{½}$ 2.14×10^6 y. It is produced in nuclear reactors (Ch. 19) through the reactions

$$^{238}\mathrm{U}(\mathrm{n},2\mathrm{n})^{237}\mathrm{U} \xrightarrow[6.75\ \mathrm{d}]{\beta^-} {}^{237}\mathrm{Np}(\approx 70\%) \tag{14.5}$$

$$^{235}\mathrm{U}(\mathrm{n},\gamma)^{236}\mathrm{U}(\mathrm{n},\gamma)^{237}\mathrm{U} \xrightarrow[6.75\ \mathrm{d}]{\beta^-} {}^{237}\mathrm{Np}(\approx 30\%) \tag{14.6}$$

14.2.5.2. Occurrence and production

About 0.5 kg ^{237}Np is contained in each ton of spent reactor fuel. By separation processes, similar to those described in Chapter 21, hundreds of kg of neptunium has been made available. Though ^{237}Np has too short a half-life to still exist naturally on earth, its optical spectrum has been observed in some stars, from which the star can be deduced to be of relatively young age ($\leq 10^7$ y, cf. §14.2.6.2 on ^{244}Pu).

14.2.5.3. Uses

So far it has only found a practical use as target material for production of ^{238}Pu. Most of the ^{238}Pu produced in this way was earlier used to power pace makers, but the main use is now as a power source (RTG) in space vehicles, used on the Mars surface (Vikings), or travelling far from the sun e.g. the two Mariner spacecraft currently leaving the solar system, or the mission to Pluto and beyond.

14.2.6. Plutonium, element 94

At the end of 1940 an isotope of element 94 (plutonium) was synthesized by Seaborg, McMillan, Kennedy, and Wahl by the bombardment of uranium with deuterons in a cyclotron:

$$^{238}_{92}\mathrm{U}(\mathrm{d},2\mathrm{n})^{238}_{93}\mathrm{Np} \xrightarrow[2.1\ \mathrm{d}]{\beta^-} {}^{238}_{94}\mathrm{Pu} \xrightarrow[88\ \mathrm{y}]{\alpha} \tag{14.7}$$

It was identified as a new element, which was distinctly different from both uranium and neptunium in its redox properties (e.g. the +3 and +4 oxidation states were more stable). A second isotope of element 94, ^{239}Pu, with a half-life of 24 000 y was isolated very shortly afterwards as the α-radioactive daughter of ^{239}Np. Although plutonium was discovered already 1941 its discovery was not revealed until 1946 due to war time issues. It was named after the now planetoid and then planet Pluto.

14.2.6.1. Isotopes

Plutonium has several rather long-lived isotopes: ^{238}Pu ($t_{½}$ 87.74 y), ^{239}Pu ($t_{½}$ 2.41×10^4 y), ^{240}Pu ($t_{½}$ 6.5×10^3 y), ^{241}Pu($t_{½}$ 14 y), ^{242}Pu ($t_{½}$ 3.73×10^5 y) and ^{244}Pu ($t_{½}$ 8.08×10^7 y). Data for all known Pu-isotopes are shown in Fig. 14.1.

14.2.6.2. Occurrence and production

Even though plutonium may have been formed in considerable amounts in the cosmic r-process (§12.5.3), the most stable isotope, ^{244}Pu, has a half-life of only 8.08×10^7 y, which prevents it from having survived on earth to our time. However, traces of ^{244}Pu have been discovered in cerium-rich rare earth minerals. If the amount found is extrapolated, considering the enrichment of plutonium in the mineral, one gram of the

earth's crust will contain 3×10^{-25} g ^{244}Pu. This would indicate about 10 g of natural plutonium remaining from the genesis of the earth, at which time it existed in parts per million of matter. ^{244}Pu decays by α-emission to ^{240}U ($t_{½}$ 14.1 h, β^-).

As discussed in §19.10, ^{239}Pu has been formed in natural uranium reactors at a later stage of the earth's evolution. Many thousands of tons of plutonium have been synthesized in commercial and military reactors; the annual global production rate in nuclear power reactors in the year 2000 was ≈1000 tons/y, contained in the spent fuel elements. The nuclear reactions and chemical separation processes are presented in Chapters 19 and 21. The build-up of heavier elements and isotopes by n-irradiation of ^{239}Pu in nuclear reactors is illustrated in Figures 14.3 and 14.4.

14.2.6.3. Uses

Experiments with ^{239}Pu confirmed theoretical predictions that it would exhibit high fissibility with both thermal and fast neutrons. This meant that ^{239}Pu in sufficient quantity and suddenly hit by a high n-flux would also experience a nuclear explosion like ^{235}U, see §14.2.4.1 above and Ch. 19. If controlled nuclear fission could be accomplished in a nuclear reactor, it would be possible to produce large amounts of plutonium by neutron bombardment of ^{238}U. The ^{239}Pu could be isolated by chemical methods which were at that time expected to be simpler than the isotopic separation required to obtain pure ^{235}U. As a consequence, the production of ^{239}Pu became a major project of the atomic bomb program of the United States during World War II; see further Ch. 19. Today, the use of Pu in nuclear explosives seems to be abandoned in favour of more easily managed highly enriched ^{235}U, or perhaps even by pure ^{233}U.

Plutonium has also a use as mixed oxide (MOX) fuel for thermal and fast nuclear reactors, see Ch 19—21. Surplus "weapons grade" ^{239}Pu metal is now often converted to PuO_2 powder for use in MOX. The lack of gamma radiation and the 87.7 y half-life of ^{238}Pu has made it a preferred energy source in radioisotope thermal electricity generators (RTG:s), *cf* §7.9.3. As mentioned in §14.2.5.3 this isotope is produced by n-irradiation of ^{237}Np. Hundreds of kilogram of $^{238}PuO_2$ has been used the space programs, some of which is now even leaving the solar system.

14.2.7. Americium and curium, elements 95 and 96

Americium and Curium were discovered by Seaborg , Ralph A. James, Leon O. Morgan and Albert Ghiorso in the USA in 1944-45. The reaction was irradiation of plutonium with neutrons in a nuclear reactor according to (14.9) and by the irradiation of plutonium with α-particles in a cyclotron according to

$$^{239}_{94}\mathrm{Pu}(\alpha,\mathrm{n})^{242}_{96}\mathrm{Cm} \xrightarrow[163\ \mathrm{d}]{\alpha} \qquad (14.8)$$

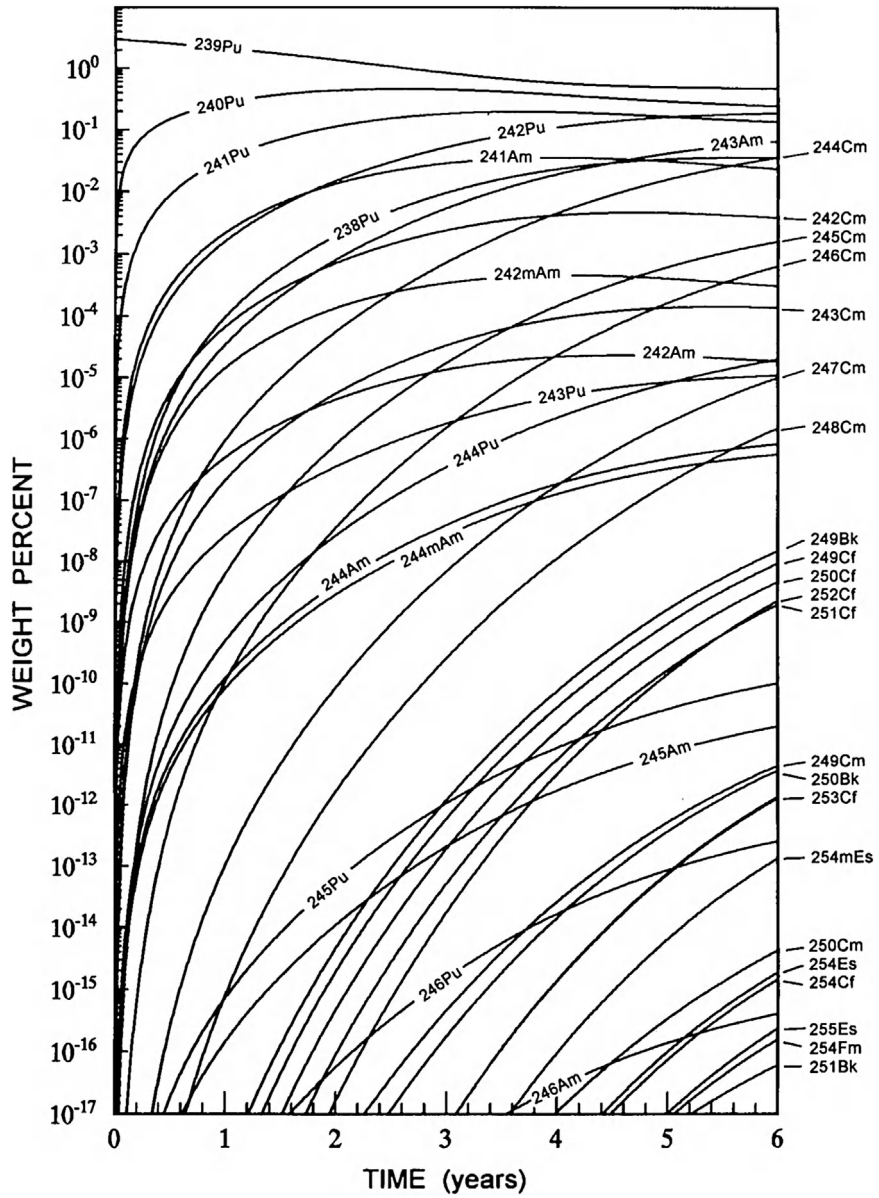


Figure 14.3 Production of higher actinide isotopes by irradiation of mixed oxide fuel (3% ^{239}Pu, rest is depleted U) at constant power in a boiling water reactor.

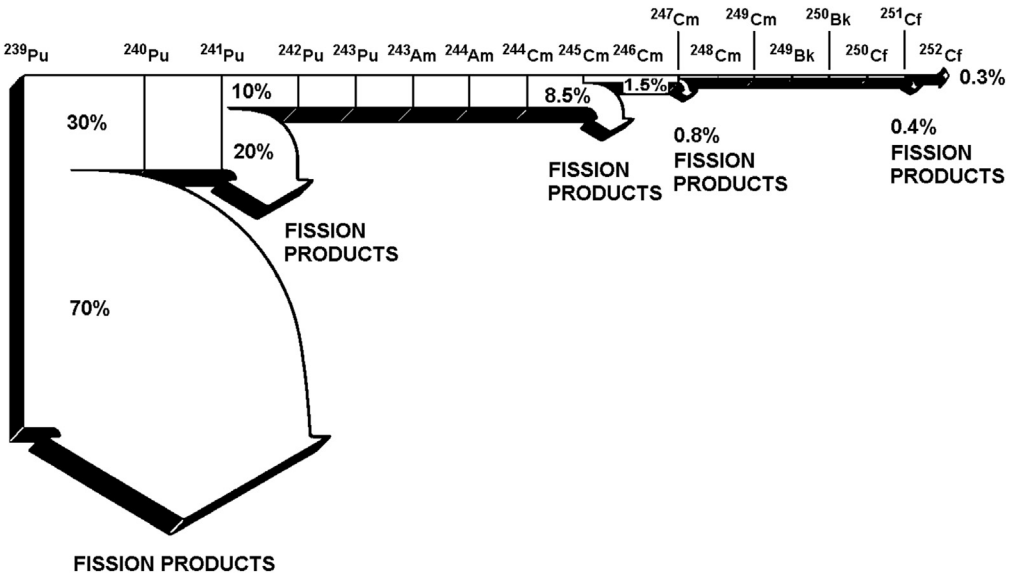


Figure 14.4 The major paths in transuranium element production in a thermal nuclear reactor; fission and n-capture compete for available nuclei *(From Seaborg)*.

14.2.7.1. Isotopes

Successive neutron capture in ^{239}Pu (as occurs in nuclear reactors) produces isotopes of plutonium which undergo β^--decay, forming transplutonium elements with atomic numbers 95 (americium) and 96 (curium) (Fig. 14.2). The reaction sequence is

$$\begin{array}{l} {}^{239}_{94}\mathrm{Pu}\,(\mathrm{n},\gamma)\,{}^{240}_{94}\mathrm{Pu}\,(\mathrm{n},\gamma)\;{}^{241}_{94}\mathrm{Pu} \\ \qquad\qquad\qquad\qquad\qquad\qquad \downarrow \beta^- \; 14.4\ \mathrm{y} \\ \qquad\qquad\qquad\qquad\qquad\qquad {}^{241}_{95}\mathrm{Am}\,(\mathrm{n},\gamma)\,{}^{242}_{95}\mathrm{Am} \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \downarrow \beta^- \; 16.01\ \mathrm{h} \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad {}^{242}_{96}\mathrm{Cm} \end{array} \tag{14.9}$$

All of the isotopes so far described have half-lives of sufficient length for ordinary chemistry although the small amounts made special laboratory techniques necessary (§18.2).

^{241}Am and ^{242}Cm also decay through spontaneous fission. The longest lived isotopes of these elements are ^{243}Am, $t_{½}$ 7370 y, and ^{247}Cm, $t_{½}$ 1.56×10^7 y, decaying by α-emission to ^{239}Np and ^{243}Pu (β^-, $t_{½}$ 5.0 h), respectively; see Fig. 14.1.

14.2.7.2. Occurrence and production

The long-lived isotopes of Am and Cm are produced in nuclear power reactors, Figs. 14.3 and 14.4. The much higher stability of the heavier nuclide ^{247}Cm (compared to ^{243}Am) can be explained by its lower Q_α (5.352 as compared to 5.638 MeV) and by the

proximity to the next magic neutron number, 152. The current world production of these element amounts to many kilograms per year. When spent fuel is reprocessed both elements are currently routed to the HLW stream. However, reprocessing methods have been developed that are able to isolate both elements in pure form in order to make it possible to try to transmute them into more short-lived or less dangerous nuclides.

14.2.7.3. Uses

The americium isotope ^{241}Am has been used as a radiation source in fire warning devices (about 30 kBq in a typical house hold device) and as a 60 keV gamma source in X-ray fluorescence analytical systems.'

No large scale use of curium exists today. Both these elements are considered among the so called *minor actinides* in used nuclear fuel. Thus, scientific work is ongoing on producing fast nuclear fuels with an appropriate content of at least americium. Thermal neutron irradiation of the mixture of curium isotopes which may be separated from spent reactor fuel initially increses the average halflife of this mixture.

14.2.8. Berkelium and californium, elements 97 and 98

By 1949 Seaborg's group had synthesized a few milligrams of $^{241}_{95}$Am from reactor bombardment of plutonium. This material was used as the target in a cyclotron bombardment. Immediately following irradiation, the target was dissolved and the products separated by passage through a column of ion exchange resin using an eluting solution of ammonium citrate (§18.2.6). An α-emitting species with a half-life of 4.5 h was identified as the isotope of element 97 of mass number 243; the name berkelium was proposed.

Later the same technique was used with a target of a few micrograms of ^{242}Cm and the first isotopes of californium ($Z = 98$) were discovered.

$$^{241}_{95}\mathrm{Am}(\alpha, 2\mathrm{n})^{243}_{97}\mathrm{Bk} \xrightarrow[4.5\ \mathrm{h}]{\mathrm{EC}} \tag{14.10}$$

$$^{242}_{96}\mathrm{Cm}(\alpha, \mathrm{n})^{245}_{98}\mathrm{Cf} \xrightarrow[44\ \mathrm{min}]{\alpha} \tag{14.11}$$

14.2.8.1. Isotopes

The longest-lived isotopes of berkelium and californium are: are ^{247}Bk, $t_{½}$ 1380 y, and ^{251}Cf, $t_{½}$ 898 y, decaying by α-emission to ^{243}Am and ^{247}Cm, respectively, see also Figure 14.1 for other isotopes of these elements.

14.2.8.2. Occurrence and production

Neither berkelium nor californium is found naturally on earth. Both are present in small amounts in fresh spent nuclear fuel. Small amounts have also spread into the environment

after atmospheric nuclear weapons tests, from a few severe reactor accidents like Chernobyl, and from releases of waste from the nuclear fuel cycle. However, current concentrations in nature of these elements are very small.

Berkelium and californium have also been produced in small amounts by n-irradiation of targets in high-flux reactors.

14.2.8.3. Uses

$^{252}_{92}$Cf has been used in small n-sources, for calibration of high energy particle detectors, and as targets for production of so called super heavy elements (SHE), but no practical use of berkelium has been found.

14.2.9. Einsteinium and fermium, elements 99 and 100

In 1952 the United States set off the first test thermonuclear explosion (code name "Mike") at the Eniwetok Island in the Pacific. The early analysis of debris from Mike by Albert Ghiorso and his team showed that a heavy isotope of plutonium, ^{244}Pu, had been made by multi-neutron capture in ^{238}U which had been part of the device. More extensive chemical purification of some of the radioactive coral from the test site proved that isotopes of elements 99 (einsteinium) and 100 (fermium) had been made in the explosion. The neutron flux during the very brief burning time of Mike had been so intense that it resulted in capture by ^{238}U nuclei of as many as 17 neutrons; see Fig. 14.5. This multi-neutron capture ended as the device blew apart and was followed by a sequence of β^--decays.

Within a year, element 99 was again synthesized by cyclotron bombardment of uranium with nitrogen

$$^{238}\mathrm{U} + {}^{14}\mathrm{N}^{6+} \rightarrow {}_{99}\mathrm{Es} \tag{14.12}$$

Several months later, Fm was isolated as a product of reactor irradiation. In this case the neutron capture occurs over a long time and β-decay processes compete with neutron capture depending on the $t_{½}$ (β^-) and the neutron flux. The reaction sequences are shown as the shaded area in Figure 14.4. Symbolically, for example the sequence ^{239}Pu+3n, β^-,+8n, 2β^-, +4n, β^-, +n, β^-, +n produces fermium. See also Figures 14.3 and 14.4 for the production of various nuclides in the sequence as a function of time for a reactor with a predominantly thermal neutron flux.

14.2.10. Mendelevium, nobelium and lawrencium, elements 101 – 103

By 1955 reactor transmutation of ^{239}Pu had produced 10^9 atoms of ^{253}Es. These atoms were deposited on gold and bombarded with helium ions to make the element 101 by the reaction

$$^{253}_{99}\mathrm{Es} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{256}_{101}\mathrm{Md} + {}^{1}_{0}\mathrm{n} \tag{14.13}$$

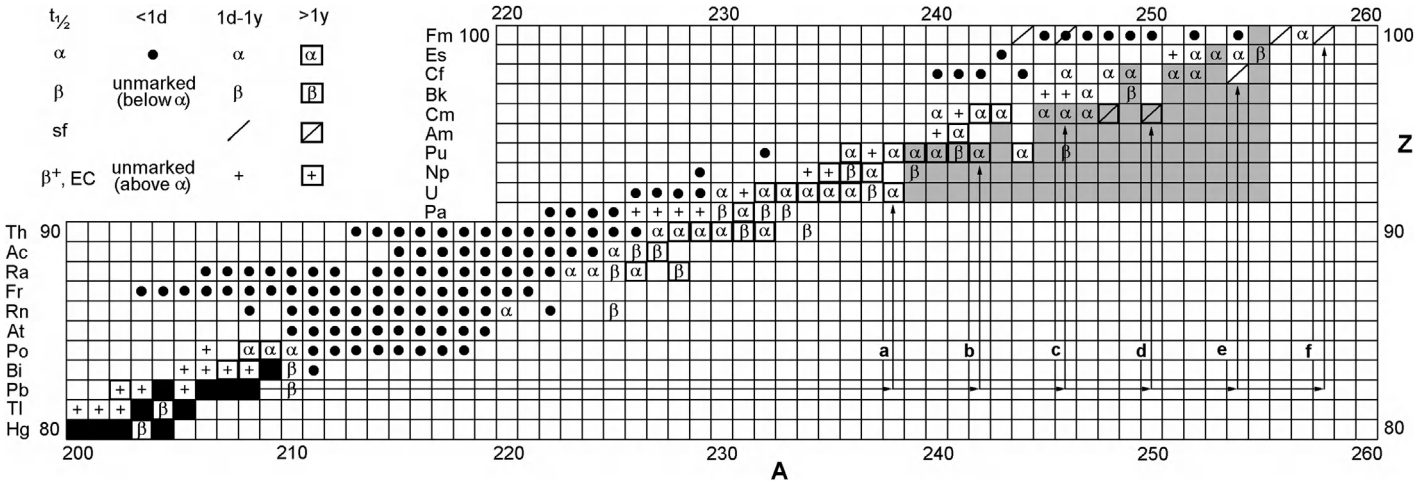


Figure 14.5 The build-up of heavy actinides through rapid multiple n-capture in lead (arrows) and in uranium (shaded area).

A new technique was used that allowed the atoms of ^{256}Md to recoil from the very thin target onto a "catcher" foil. Thirteen atoms of ^{256}Md ($t_{½}$ 1.3 h) were made in 9 h of irradiation and isolated by rapid elution from a column of cation exchange resin using a solution of α-hydroxy isobutyric acid. The elution showed 5 atoms of element 101 (identified by the spontaneous fission of the daughter fermium) and 8 atoms of the fermium daughter (Fig. 14.8; eqn. (14.14)). The recoil – ion exchange technique used in identification of element 101 by Thompson, Choppin, Seaborg, Harvey and Ghiorso has been successfully applied in the discovery of other heavier elements.

$$^{256}_{101}\mathrm{Md} \xrightarrow[1.3\ \mathrm{h}]{\mathrm{EC}} {}^{256}_{100}\mathrm{Fm} \xrightarrow[2.6\ \mathrm{h}]{\mathrm{sf}} \tag{14.14}$$

So far we have described two methods of production of heavy elements: n-irradiation followed by β^--decay, and accelerator induced reactions according to relations 14.8—14.13. Multiple n-capture has its limitation in the successively decreasing yields, while α-reactions usually yields product nuclides further away from the bottom of the *Z*-versus-*N* stability valley (cf. e.g. Fig. 4.1, 14.1 and 14.6) resulting in shorter half-lives. Heavier isotopes (i.e. isotopes with larger mass numbers, that is more neutron rich) of the product element are expected to be more stable (i.e have longer half-lives). However, by irradiating the heavy target atom by "heavy ions", e.g. cromium ions, three advantages can be achieved in one reaction: the charge of the compound nucleus is increased

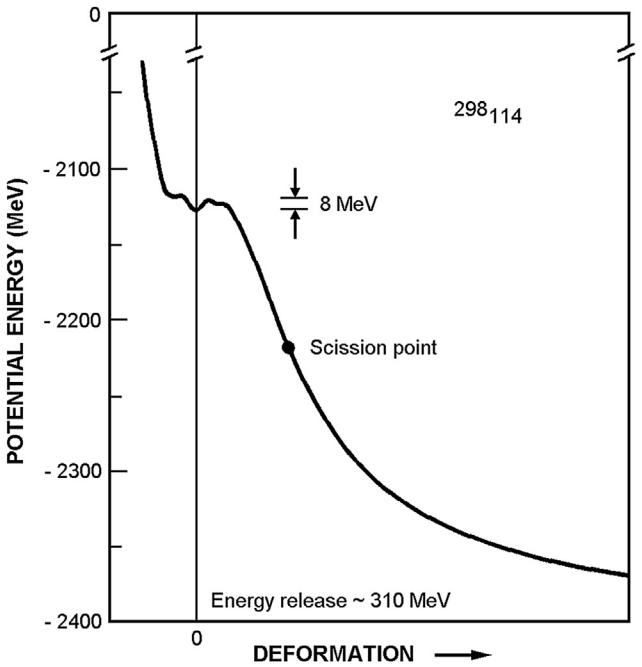


Figure 14.6 Dependence of potential energy on deformation for a nucleus like $^{298}_{114}$. *(From R. Nix.).*

several steps, while at the same time more neutrons ("glue") are added, and, if the reaction energy is negative, giving a less exited compound nucleus, hopefully leading to a much heavier and more stable product. This method is used today for the production of the heaviest actinide elements and of transactinide elements.

In 1957 Russian scientists at the Joint Institute for Nuclear Research (JINR) at Dubna (now Flerov Laboratory for Nuclear Reactions, FLNR) claimed to have synthesized element 102 by irradiating ^{241}Pu with ^{16}O ions, using nuclear emulsions for α-energy determination.

$$^{241}_{94}\mathrm{Pu} + {}^{16}_{8}\mathrm{O} \rightarrow {}^{252}_{102}\mathrm{No} + 5{}^{1}_{0}\mathrm{n} \tag{14.15}$$

They assigned an 8.8 ± 0.5 MeV α-decay with a half-life of 2−40 s, to element 102. However, their results have been contested by scientists at the Lawrence Berkeley Laboratory (LBL) in California, who produced and identified element 102 in 1958 by use of a double-recoil technique[1]. The reaction scheme was

$$\begin{array}{l} {}^{246}_{96}\mathrm{Cm} + {}^{12}_{6}\mathrm{C} \xrightarrow{\text{1 st recoil}} {}^{254}_{102}\mathrm{No} + 4{}^{1}_{0}\mathrm{n} \\ \qquad\qquad\qquad\qquad\quad \alpha \downarrow \text{2nd recoil} \\ \qquad\qquad\qquad\qquad\quad {}^{250}_{100}\mathrm{Fm} \end{array} \tag{14.16}$$

Lawrencium, element 103, was synthesized at Berkeley in 1961 by the reactions

$$^{250,1,2}_{98}\mathrm{Cf} + {}^{10,11}_{5}\mathrm{B} \rightarrow {}^{258}_{103}\mathrm{Lr} + (2-5){}^{1}_{0}\mathrm{n} \tag{14.17}$$

using a system where the recoiling products were caught on a moving band, which rapidly transported them to an array of energy sensitive solid state detectors (cf. Fig. 17.3). The half-life of the product (later determined as ∼4s) was too short to allow any chemistry. It was the first actinide element to be identified through purely instrumental methods.

14.2.11. The transactinide elements, Z ≥ 104

The transactinide elements, some of which are often also called superheavy element (SHE), are the elements beyond the end of the actinide series, i.e. Z > 103. Many have a nucleus that survives only because of microscopic shell stabilization. When the number of protons increases the repulsive forces between the protons increase, and the number of chemical elements is therefore limited by the increasing probability for spontaneous fission. This macroscopic behavior is governed by shell effects. Nuclei can survive beyond the macroscopic limit, where the necessary balance between the attractive

[1] The name *nobelium* was selected to honor the Swedish industrialist Alfred Nobel, who donated his fortune to be used as prizes to those who have contributed most to progress in the fields of science, the humanities and peace.

nuclear force and the repulsive Coulomb force is achieved only through shell stabilization. The number of protons where this occurs is around Z = 104. Studies of their properties are linked to an understanding of the physics and chemistry at the end of the periodic table. Due to conflicting claims of discovery there have been several names used for some of these heavy elements, see Table 14.7.

According to calculations made e.g. by Nilsson et al., Scharff-Goldhaber, Strutinsky and Swiatecki in the late 1960:ies there may also be an area near the next (predicted) double shell closure above lead where nuclei may have surprisingly long half-lives. The first calculations indicated half-lives of millions of years or even stability. Today the predicted half-lives are a few orders of magnitude shorter but still very long. On the way to this "island of stability" in the "sea of instability" regions with deformed nuclei exhibiting a varying degree of stronger binding have also been predicted.

Some years ago it was generally assumed that the next proton magic number after 82 would be 126 since this was the neutron pattern. However, more elaborate theoretical studies indicate that the proton shell closure occurs for Z = 114 (or possibly 112). The most probable neutron magic number is 184. Taking into account also pairing effects, relativistic theory predicts additional shell closures around Z ~ 108–110 as well as N ~ 162 and possibly N ~ 172.

Experiments show evidence of enhanced nuclear stability around N ~ 162 and possibly in the region around N ~ 172. So far N = 184 has not been reached but experimental data suggest that stability increases with increasing N in the region N = 162–177, with N = 177 as the highest neutron number reached so far.

In 2012 elements up to 118 have been artificially synthesized (only the elements up to and including 112, as well as 114 and 116 are, however, so far acknowledged by IUPAC) and efforts are being made to create Z = 119 and 120 in the laboratory.

14.2.11.1. Synthesis of elements with Z ≥ 104

The only known way to produce elements with $Z > 103$ is heavy–ion reactions, i.e. fusion of two heavy nuclei. The compound nucleus (CN) is then de-excited through neutron evaporation. The formed nuclide decays by successive alpha decays. For each step in the neutron evaporation process, there is a strong competition between spontaneous fission and alpha decay that makes these heavy systems highly fissile. The survival probability of the compound nucleus is only of the order of 10^{-2} to 10^{-3} per evaporation step. Table 14.8 lists the nuclear reactions used for the first production and identification of the very heavy elements.

The daughter products of the CN are identified by genetic correlations and simultaneous lifetime measurements. Alpha decay half-lives from some seconds and down to microseconds and rapidly decreasing cross-sections for every additional proton limits the study of their chemistry and physics to 'single atoms', which makes experimental studies extremely difficult.

Two types of reaction mechanisms are in use for synthesis of the heaviest elements. "Cold" fusion reactions are performed with lead or bismuth targets that are bombarded with appropriate projectiles. One or two neutrons are evaporated from the CN. The daughter products are typically formed via successive alpha-decays ending by spontaneous fission (SF) in the region of the known elements. They are identified via parent-daughter correlations or the chemical properties of the final decay product. Cold fusion reactions have been used to produce elements 107 to 112 at GSI in Darmstadt. An estimate of the limit of cold fusion experiments comes from the synthesis of a single atom of Element 113 at RIKEN. The production cross-section was on the order of femto-barns, which makes it unlikely that cold fusion can be used to produce even heavier elements.

"Hot" fusion reactions are performed with an actinide target and an appropriate, lighter projectile. The reaction is followed by the evaporation of 3 to 5 neutrons. Hot fusion reactions have been used successfully to produce the more neutron rich nuclides from rutherfordium to element 118. These decay chains often end in unknown regions making an unambiguous identification of the nuclides challenging. The elements 113 to 118 have been produced with "hot" fusion at FLNR in Dubna using actinide targets from uranium to californium with beams of the extremely rare doubly magic isotope ^{48}Ca.

The short half-lives and the small amounts of product did not allow chemical identification of the transactinide elements in the initial experiments. Identification by nuclear properties used to be less conclusive as it usually depends on correlation with predicted energies and half-lives obtained by extrapolation of systematic trends. The controversies resulting from the uncertainties inherent in this approach have been a feature of the history of the light transactinide elements, just as they were for nobelium and lawrencium. In fact, it is still uncertain which claims represent the actual discovery for some of these elements, the Russian group at Dubna or the American group at Berkeley. Both groups undoubtedly have produced isotopes of elements 104–106. Partially the disagreements are due to the different techniques used. At both laboratories nuclear reactions using hot fusion, were used for the first synthesis of elements 104–105, see Table 14.7. This type of reaction was used at Berkeley also for production of element 106, while the Dubna group utilized cold fusion.

During the last 15 years hot fusion reactions with ^{48}Ca as projectile have been used in Dubna to synthesize the elements 113-118. Identification has been made by genetic correlations. A problem in the identification is, however, that all decay chains end with previously unknown spontaneously fissioning nuclides of e.g. dubnium or seaborgium. Thus, there is a need for chemistry experiments to show that these end members of the decay chains follow the expected chemical behavior. To connect the isolated "Dubna island" to the known region of the chart of nuclides is therefore a challenging task. Until now, IUPAC has accepted the discovery of elements 114 and 116 because they have been confirmed in experiments at Berkeley but not 113, 115, 117 and 118.

The predictions that the half-lives of some heavy elements may be very long have led to searches for evidence of their existence in nature. Several reports of tentative evidence for such superheavy elements have appeared but subsequent experimentation has explained the observations as based on known elements with only a few exceptions.

14.3. ACTINIDE PROPERTIES

14.3.1. The actinide series

As early as 1923 N. Bohr suggested that there might exist a group of 15 elements at the end of the Periodic Table that would be analogous in their properties to the 15 lanthanide ("rare earth") elements. This idea, combined with the increasing stability of the +3 oxidation state for the transuranium elements as the atomic number increases from $Z = 93$ to 96, led Seaborg to the conclusion that these new elements constituted a second rare earth series whose initial member was actinium. As the atomic number increases from 90, electrons are added in the 5f subshell similar to the occupation of the 4f subshell in the lanthanides, see Table 14.3. This series would be terminated with element 103 since this would correspond to the addition of 14 electrons for a completed 5f subshell.

Table 14.3 Electronic configuration, radii (in picometres, pm), and oxidation states in (acid, non-complexing) aqueous solution

Atomic number	Element	Metallic radius (pm)	Atomic (g)† config.	Effective ionic radius†		Oxidation* states
				M^{3+}	M^{4+}	
89	Ac	188	$5f^0 6d 7s^2$	111.9		**3**
90	Th	180	$5f^0 6d^2 7s^2$	(108)	97.2	(3) **4**
91	Pa	163	$5f^2 6d 7s^2$	(105)	93.	(3) 4 **5**
92	U	156	$5f^3 6d 7s^2$	104.1	91.8	3 4 5 **6**
93	Np	155	$5f^4 6d 7s^2$	101.7	90.3	3 4 **5** 6 (7)
94	Pu	160	$5f^6 7s^2$	99.7	88.7	3 **4** 5 6 (7)
95	Am	174	$5f^7 7s^2$	98.2	87.8	**3** 4 5 6
96	Cm	175	$5f^7 6d 7s^2$	97.0	87.1	**3** 4
97	Bk		$5f^9 7s^2$	94.9	86.0	**3** 4
98	Cf		$5f^{10} 7s^2$	93.4	85.1	(2) **3**
99	Es		$5f^{11} 7s^2$	92.5		(2) **3**
100	Fm		$5f^{12} 7s^2$			(2) **3**
101	Md		$(5f^{13} 7s^2)$	89.6		2 **3**
102	No		$(5f^{14} 7s^2)$			**2** 3
103	Lr		$(5f^{14} 6d 7s^2)$	88.2		**3**

*The most stable oxidation state in (acid, non-complexing) aqueous solution is given in bold, most unstable (or only found in the solid state) within parenthesis.
†Electronic configuration and ionic radii (coordination number 8) according to Katz, Seaborg and Morss and to Seaborg and Loveland.